

Homework

1. Show that $\mathbb{Z}_6$ is a ring with unity under addition and multiplication.
2. Show that the set $\sqrt{2}\mathbb{Z} = \{x: x = \sqrt{2}n \text{ for some } n \in \mathbb{Z}\}$ is a ring.
3. Let F be the set of all functions from $\mathbb{R}$ to $\mathbb{R}$. Define addition and multiplication as

$$(f + g)(x) = f(x) + g(x) \text{ and } (fg)(x) = f(x)g(x).$$

Show that F is a commutative ring with identity.

4. Prove that the ring above, F , is not an integral domain.

Hint Here are two *indicator functions*

$$\mathbf{1}_{[0,\infty)}(x) = \begin{cases} 1 & \text{if } x \in [0, \infty) \\ 0 & \text{if } x \notin [0, \infty) \end{cases} \text{ and } \mathbf{1}_{(-\infty,0]}(x) = \begin{cases} 1 & \text{if } x \in (-\infty, 0] \\ 0 & \text{if } x \notin (-\infty, 0] \end{cases}$$

The indicator function is a function that has an output of either 1 or 0. For example, $\mathbf{1}_{[0,\infty)}(2) = 1$ and $\mathbf{1}_{[0,\infty)}(-2) = 0$. Now consider the functions $f(x) = x\mathbf{1}_{[0,\infty)}$ and $g(x) = x\mathbf{1}_{(-\infty,0]}$.

5. Let R and S be rings. Define addition and multiplication on the Cartesian product $R \times S$ by,

$$(r, s) + (r', s') = (r + r', s + s') \text{ and } (r, s)(r', s') = (rr', ss')$$

for $r, r' \in R$ and $s, s' \in S$. Prove that $R \times S$ is a ring.

6. Show that the set $k\mathbb{Z} = \{x: x = kn \text{ for some } n \in \mathbb{Z}\}$ for any integer k is a subring of $\mathbb{Z}$.
7. Prove that $\mathbb{Q}$ is an integral domain.
8. Prove that $\mathbb{Z}_6$ is not an integral domain.
9. Let a and b be two nonzero elements in a ring R . Prove that in general, $ax = b$ does not have a unique solution.

Hint It should be understood that x must be in R . An example is all that is needed. You can find an example that has two solutions, or find an example with no solution.

10. Prove that $\mathbb{Z}_n$ is not an integral domain (under the usual addition and multiplication) if n is not a prime (n is assumed to be a positive integer).

Hint If n is not prime then $n = mq$ for two integers $1 < m, q < n$. Looking at the example seen in exercise might help.

Presentation Problems

1. Prove that the intersection of any two subrings of a ring R is also a subring of R .
2. Give an example of a ring with elements a and b with properties that $ab = 0$ but $ba \neq 0$.
3. Let m and n be positive integers with least common multiple k . Prove that $m\mathbb{Z} \cap n\mathbb{Z} = k\mathbb{Z}$.
4. Prove that a ring that is cyclic under addition is commutative.
5. Prove that if a ring element has a multiplicative inverse, it is unique.
6. Prove that a field is an integral domain.
7. Prove that $\mathbb{Z}_p$ is an integral domain if p is prime (you can assume that it is a commutative ring with identity).
8. Prove that $\mathbb{Z}_p$ is a field.